Vamshi Muluguri

Hyderabad, India

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Summary

AI Engineer with hands-on experience building enterprise agentic AI, RAG, and workflow-automation solutions using Python, FastAPI, LangChain, and LangGraph. Experienced in reusable agent capabilities, LLM tool calling, retrieval with Azure AI Search, human-in-the-loop controls, evaluation, observability, and production deployment on Azure Kubernetes Service (AKS). Focused on secure, reliable, maintainable AI services for enterprise workflows.

Skills

Languages & Backend: Python, FastAPI, Java, Pydantic, REST APIs, Async Python
Agentic AI & Generative AI: LangChain, LangGraph, Agentic Workflows, RAG, LLM Tool Calling, Structured Outputs, Guardrails, Prompt & Context Engineering, Azure OpenAI, Azure AI Search, DeepEval
Data: PostgreSQL, SQLAlchemy, Alembic, ChromaDB, FAISS, Azure Blob Storage
Development Quality: Pytest, Automated Testing, Test-Case Design, Code Review Workflows
AI-Assisted Development: Experience with AI-powered coding assistants such as Claude Code, OpenAI Codex, and GitHub Copilot
Cloud/DevOps: Azure, Docker, Kubernetes (AKS), Helm, CI/CD, OpenTelemetry, Grafana
Security & Agent Platforms: Identity & Access Control, Tool Authorization, Agent Memory/Context Management, mTLS, Vault/Key Vault, Human Approval Workflows, Dry-Run Controls, Production Releases, Rollback & Hotfix Handling, Environment Promotion (Dev/UAT/Prod)

Professional Experience

UBS **Oct 2025 – Present**
AI Engineer *Hyderabad, India*

- Implemented an **AI-powered Request Management Service** using Python and FastAPI to automate triage, enrichment, and resolution of enterprise service requests across Atlas and ServiceNow.
- Engineered **LangChain/LangGraph agentic workflows** with **RAG** over Confluence and **Azure AI Search**, reaching a **32% AI-enriched acceptance rate** and approximately **30% less manual triage time**.
- Built a reusable **Agent Skills framework** that generates, evaluates, and exports executable skills from knowledge articles, with a self-onboarding frontend to standardize how teams add use cases.
- Integrated **human-in-the-loop approval** through an MS Teams bot, with auditable automation and dry-run mode for controlled execution of consequential actions; emphasized explicit tool-use controls and approval checkpoints.
- Applied evaluation-driven development with **DeepEval**, added **OpenTelemetry/Grafana** observability, and supported test-case design to improve evaluation quality and operational visibility.
- Containerized services with Docker and deployed to **AKS** using Helm, integrating PostgreSQL, Vault/Key Vault secrets, and mTLS.

Edunet Foundation **Feb 2024 – Jun 2024**
Full Stack Developer Intern *Remote*

- Built and deployed a blogging platform with a responsive **React.js** interface to streamline the publishing workflow for trainers.
- Implemented **JWT**-based role access and **Zod** validation to standardize and validate full-stack data.

Projects

Agentic CI/CD Test Failure Root Cause Analyzer
FastAPI, LangGraph, Azure OpenAI, RAG, ChromaDB, React, GitLab MCP

- Built a LangGraph multi-agent workflow for log parsing, failure classification, and context-aware fix recommendations for GitLab CI/CD failures.
- Implemented an MCP-powered similar-case finder using ChromaDB/FAISS to retrieve prior incidents and validated fixes, surfaced through GitLab comments, Teams, and a React dashboard.

NutriScan AI – Nutrition Analysis Platform
LangGraph, Azure OpenAI Vision, RAG, ChromaDB, FastAPI, ReAct

- Designed a five-agent workflow using vision, OCR, and RAG for ingredient extraction, nutrition scoring, allergy detection, and personalized recommendations.
- Implemented conditional routing, retries, fallback handling, JWT authentication, and Docker Compose deployment.

Education

Sri Indu College of Engineering and Technology **Hyderabad, India**
B.Tech in Computer Science and Engineering *2019 – 2023 | CGPA: 7.47*